

MSc PROJECT DEFINITION 2021-22

This project definition must be undertaken in consultation with your supervisor. The feasibility of the project should have been assessed and the project aims should be clearly defined.

Submission of this document implies that you have discussed the specification with your supervisor.

Project Title: Creating Procedural Generated Survival Game

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PROJECT AIMS:

The aim is to take advantage of the method 'procedural generation' to create data using algorithms. Specially, when talking about computer games, automatically generated worlds and features is a technique extensively used without the need of developers to build them for eg: Minecraft, No Man's Sky, etc.

Using the same technique and since this topic has been a part of my module too for this semester, I plan to create open worlds for a survival mode given the time frame for the submission of this project I'll be able to create few levels/a random open world with different features in each one of them.

The goal is to create a survival video game where the user/player is able to interact with the elements in the open world and use resources to interact around. The result would be achieved by generation of optimised terrains and resources distributed randomly throughout the open world in the game.

PROJECT OBJECTIVES:

- Modelling, texturing and animation of the main character in of the game
- Scripting the movements of the main character
- Procedural Generation of terrains
 - Algorithms
 - Noise
 - Map
 - Mesh
 - Detailing
 - Water
- Generation of resources
- Scripting the character's interaction with the environment
 - Collecting resources
 - Crafting/buying/selling
- Storing Memory

METHODOLOGY:

- Using the free open source tool called 'Blender' I'll be creating model of the main character and add animation and textures.
- I would be using available assets on the Unity Store for standard movement packages and editing them if required.
- Using noise algorithm I'll be generating random input data such as clouds, textures, landscapes, etc.
- Obtaining the 2D noise map to begin the generation of terrains.
- From the noise map I'll generate a 3D mesh for terrains.
- A falloff map which is basically a 2D texture with a gradient will be used to generate island-shaped terrain surrounded by water.
- Using Raycast I'll be generating resources that the player can interact with and find in the open world.
- Using script based on Raycast, I'll programme the interaction of player with the resources and terrain in the open world.

PROJECT MILESTONES:

- Creating the base of a survival game using Unity
- Using procedural techniques successfully achieving generated world
- Optimising the game to be playable by all users with minimum computational resources.
- Bases on the maps generated make in-game resources available for the player using the same method of procedural generation.
- Make the game look appealing which includes background, terrain, animations, music effects, etc.

REQUIRED KNOWLEDGE/ SKILLS/TOOLS/RESOURCES:

- Unity 3D - Game Engine
- Blender - Animations
- Visual Studio - C# Scripting
- Photoshop - Basic editing

TIME-PLAN:

Week 1 starts from April 1, 2022

Gantt Chart

